

CARLOS VICENTE SÁEZ RUFETE

COMPUTER VISION ENGINEER | ML ENGINEER | MLOPS

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PROFILE

Telecommunications engineer working in industrial computer vision at AUTIS Ingenieros since 2019, currently leading a technical team. Development of object detection (YOLO), image classification and anomaly detection (EfficientAD) models, optimisation with ONNX and TensorRT for GPU inference, and training on Kubernetes with GitLab CI/CD. MSc in Advanced and Applied Artificial Intelligence, Universitat de València (2024-2025).

WORK EXPERIENCE

Computer Vision Engineer (Team Lead)

AUTIS Ingenieros, S.L.U. | May 2024 - Present

- Training of object detection models with YOLO (Ultralytics) for industrial inspection.
- Image classification models and unsupervised anomaly detection with EfficientAD.
- Model export to ONNX and optimisation with TensorRT (.engine) for GPU inference, integrated through a DLL.
- Model training on Docker and Kubernetes and maintenance of CI/CD pipelines in GitLab.
- Integration of vision systems into high-availability industrial production lines.
- Technical coordination of the team, resource management and project planning.

Computer Vision Engineer (Senior)

AUTIS Ingenieros, S.L.U. | January 2022 - May 2024

- Analysis of the industrial environment: repeatability, lighting and interference.
- Digital image processing, optimisation and hardening of the vision pipeline.
- Test automation and model validation.
- Backend and frontend development of computer vision solutions.
- Resource and project management and advanced technical support.

Computer Vision Engineer (Junior)

AUTIS Ingenieros, S.L.U. | February 2019 - January 2022

- Laboratory testing and analysis of vision hardware.
- Image processing and algorithm testing.
- Commissioning and support in industrial environments.

PROJECTS

Object detection with YOLO 2025 - present | Industry

Fine-tuning of YOLO models to detect surface defects, with training on Docker and Kubernetes. Put into production as a TensorRT engine (.engine), with GPU inference integrated through a DLL.

YOLO, TensorRT, Python, Docker, Kubernetes, GitLab

Paint defect classification Automotive

Filtering and classification of detected paint defects: training of filtering and classification models and putting them into production.

Python, TensorFlow, Scikit-learn, Kubernetes, Docker, TensorRT, SQL

Robotic paint defect inspection 2023 - present | Automotive

Commissioning of robotic systems that detect paint defects on plastic parts.

Robotics, industrial vision, LabVIEW

Robot path correction with 3D vision 2019 - 2021 | Automotive

Correction of robot paths from stereo camera point clouds, with part registration using ICP. Also applied to robotic door assembly with real-time guidance.

Stereo vision, point clouds, ICP

Paint defect inspection systems 2021 - 2023 | Automotive

Commissioning of systems that detect paint defects on car bodies.

Industrial vision, LabVIEW

Washing machine screwdriving Industry

Detection of screws on washing machines for screwdriving projects, using pattern matching and traditional computer vision.

Hardware selection, LabVIEW

OCR on glass **2020 | Industry**

Optical character recognition (OCR) on glass.

Hardware selection, LabVIEW, traditional vision

Connector poka-yoke on cooktop boards **2020 | Industry**

Detection of whether connectors are present on the integrated electronic boards of cooktops, using pattern matching.

Hardware selection, LabVIEW, traditional vision

EDUCATION

MSc in Advanced and Applied Artificial Intelligence

Universitat de València

2024 - 2025

BEng in Telecommunications Engineering, Sound and Image

Universitat Politècnica de València

2014 - 2018

SKILLS

Languages: Python, SQL

Deep learning and vision: PyTorch, TensorFlow, Ultralytics (YOLO), OpenCV, Scikit-learn, Scikit-image

Classical and 3D vision: OCR, pattern matching, stereo vision, point clouds, ICP, vision hardware selection

MLOps and inference: ONNX, TensorRT, Docker, Kubernetes, GitLab CI/CD, MLflow

Tools: Git, VS Code, LabVIEW

Spoken languages: Spanish and Valencian (native), English (advanced)